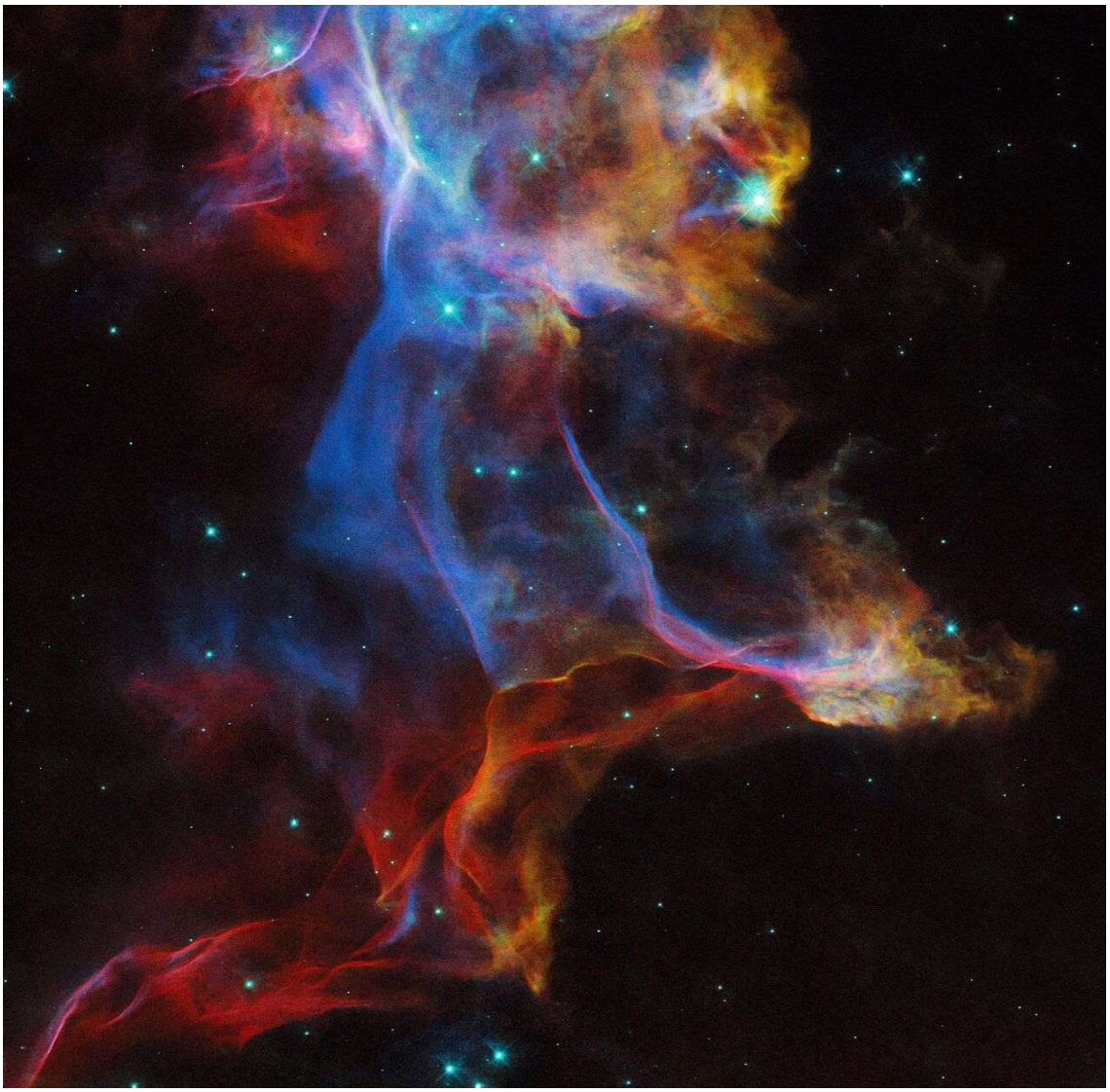
Are We Alone in the Universe?

-Aniket



NASA/ESA Hubble Space Telescope image a supernova remnant called the Veil Nebula.

Introduction

Life on Earth exists because of a remarkable combination of circumstances. Our planet sits at just the right distance from the Sun for liquid water to exist. It possesses a stable atmosphere rich in nitrogen and oxygen, a protective magnetic field, and the chemical ingredients necessary for life.

The emergence of life required warmth, water, energy, and a delicate mixture of gases and elements working together over billions of years.

At first glance, this may seem like an extraordinary coincidence. Yet when we look out into the cosmos, we discover that Earth is not unique in every respect. Astronomers have identified thousands of planets orbiting distant stars, and many of them appear to be rocky worlds similar in size to Earth. If entire course of life on Earth is a remarkable co incidence, then it is only logical to think that similar circumstances are occurring or have already occurred in the vast expanse of the universe. And there are many such discovered “Earth like” planets. Some even lie within their star's habitable zone, where temperatures may allow liquid water to exist. A number of these planets may even possess atmospheres and chemical compositions broadly similar to our own.

But that does not entirely answer our question. Detection of habitable planet is one work and finding concrete evidence of life in such exoplanets is another. One logical prediction that can made, looking at the Earth analogs is that life if exists in such regions will be carbon based. In other words there is very good chance that aliens can be “flesh and blood” just like us. If the planet is “Earth like”, it only makes sense that the possible intelligent habitants are “Human like”. But scientists are still far away from any evidence suggesting such life forms.

The Scale of the Cosmos

As of 2026, astronomers have confirmed more than 6,000 exoplanets (planets orbiting other stars), and thousands more candidates are awaiting confirmation. Some of these exoplanets are considered habitable zones and fir for earth like ecosystems.

Kepler-186f was the first Earth-sized planet discovered within the habitable zone of another star, marking a major milestone in astronomy. Located approximately 500 light-years away, it is only slightly larger than Earth and likely possesses a rocky surface. The planet receives less sunlight than Earth, meaning any potential life would experience a dimmer sky. Nevertheless, its discovery provided the first strong evidence that Earth-sized planets capable of supporting liquid water are not unique to our Solar System.

Among the most promising Earth-like worlds discovered so far is **TRAPPIST-1e**, one of seven rocky planets orbiting the ultra-cool dwarf star TRAPPIST-1. Located about 40 light-years from Earth, it is remarkably similar to our planet in both size and density, suggesting a rocky composition. It receives a comparable amount of energy from its star as Earth receives from the Sun, placing it firmly within the system's habitable zone.

Proxima Centauri b holds a special place among exoplanets because it orbits Proxima Centauri, the closest star to our Solar System. At just over four light-years away, it is humanity's nearest potentially habitable neighbor. The planet is believed to be roughly Earth-sized and lies within its star's habitable zone, where temperatures may allow liquid water to exist. While its parent star is known for powerful stellar flares that could threaten any atmosphere, Proxima Centauri b remains one of the most compelling candidates for future exploration and the search for signs of alien life.



A picture of Kepler-186f, one of the possibly habitable exoplanets discovered

Other prominent names include **TOI-700 d, LP 890-9 c, Kepler -422b,** etc. They all suggest the possibility of alien life. This raises an intriguing question: if the conditions that made life possible on Earth exist elsewhere, why should Earth be the only planet to host life?

The universe contains hundreds of billions of stars in our galaxy alone, and hundreds of billions of galaxies beyond that. Even if the probability of life emerging on a suitable planet is extremely small, the sheer number of potential worlds suggests that life may have arisen many times throughout cosmic history.

Yet this leads us to one of the most famous questions in science:

“Where are they?”

If intelligent civilizations are common, we might expect to see evidence of them. We might detect radio signals, observe large-scale engineering projects, or encounter signs of technological activity. The remarkable life supporting conditions of Earth, are actually fairly detectable across the cosmos, which makes it fairly common given only a small fraction of the universe is uncovered by us so far. However, despite decades of searching, we have found no confirmed evidence of extraterrestrial civilizations. This apparent contradiction between the high probability of alien life and the lack of observed evidence is known as the **Fermi Paradox**. The Fermi Paradox is one of the most famous paradoxes in the domains of extraterrestrial biology.



The Allen Telescope Array, Hat Creek Radio Observatory.

Photo: SETI Institute/Simon Steel

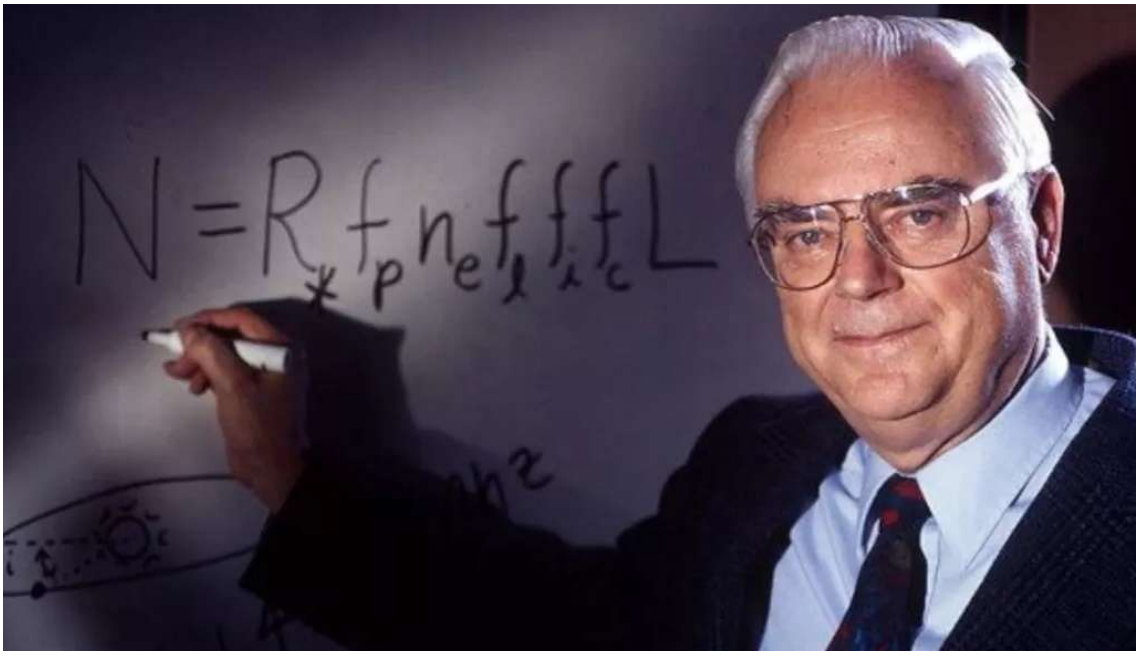
The observable universe contains hundreds of billions of galaxies, totaling up to 10^{22} stars. Many of these are older than our Sun and have Earth-like planets in their habitable zones. Even with modest, near-future rocket technology, a single advanced civilization could colonize the entire Milky Way galaxy in a few million years. This is a tiny fraction of the galaxy's 13-billion-year history. Despite humanity's expansion and astronomical surveys (like SETI), we have not observed any megastructures, radio broadcasts, or spacecraft. All this information suggests an inter planetary interaction in the course of human history, yet there is a lack of evidence supporting so. It feels almost impossible to believe that despite all clue humans have found almost nothing, it feels like we are the only intelligent species in this universe. Three most popular potential explanations of the paradox exist.

The Great Filter is a hypothesis that prevents the civilizations from reaching interstellar communications. The “filter” could be behind us (meaning abiogenesis—going from non-living to living matter—is incredibly rare) or ahead of us (meaning advanced civilizations inevitably destroy themselves with technology or war).

They Are Here but We Can't See Them-Some theories propose that advanced civilizations are aware of us but choose not to interact. This includes the **Zoo Hypothesis** (they treat Earth like a protected wildlife preserve) or the **Planetarium Hypothesis** (we are isolated in an engineered simulation)

They Are Hiding (The Dark Forest Theory) This unsettling idea posits that space is hostile, and since the first civilization to strike could dominate, the safest bet for any intelligent species is total silence. To avoid destruction, advanced civilizations deliberately conceal their locations and emissions.

Although each explanation feels realistic, they are mere hypothesis and lacks a rock-solid proof of its truth. Maybe the real clues are still hidden, or are already unraveled but overlooked. These three possibilities are sometimes mathematically expressed as circumstantial outcomes of what scientists call the Drake equation. The Drake equation provides us a comprehensive idea of the existence of interstellar life based on all the clues available so far. The equation calculates the number of detectable civilizations (N) by multiplying seven variables



Frank Donald Drake (1930 – 2022) was an American astrophysicist and astrobiologist. He proposed an equation suggesting a framework for scientific discussion in extraterrestrial domains which is known as the Drake equation.

“Where is everybody?”

To approach this question scientifically, astronomers often turn to the **Drake Equation**, proposed by Frank Drake in 1961. Rather than providing a direct answer, the equation estimates the number of communicative civilizations that might exist in our galaxy by multiplying together factors such as:

- The rate at which stars form.
- The fraction of stars with planets.
- The number of potentially habitable planets.
- The likelihood that life emerges.
- The probability that intelligent life develops.
- The likelihood that intelligent species create detectable technology.
- The length of time such civilizations remain detectable.

Some of these factors are becoming easier to estimate as our knowledge of exoplanets improves. Others, such as the probability of life arising from non-living matter, remain almost completely unknown.

The equation calculates the number of detectable civilizations (N) by multiplying seven variables

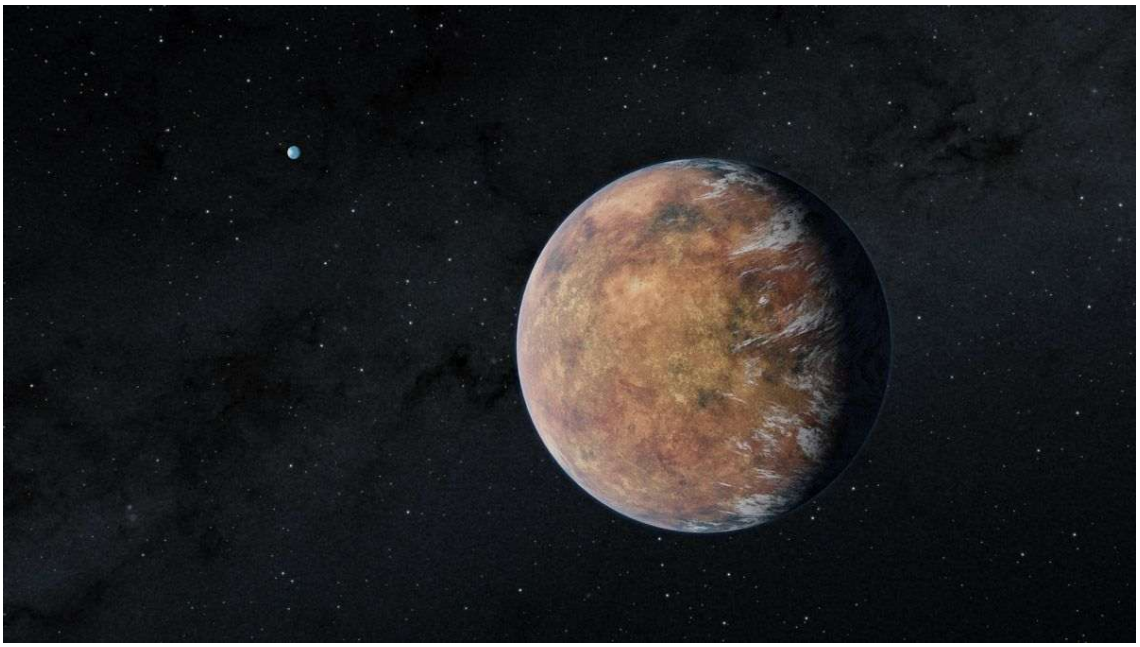
$$N = R \times f_p \times n_e \times f_l \times f_i \times f_c \times L^*$$

Where:

- **N** = The number of detectable civilizations currently existing in the Milky Way.
- **R*** = The average rate at which new stars form in our galaxy each year.
- $\mathbf{f_p}$ = The fraction of those stars that possess planetary systems.
- $\mathbf{n_e}$ = The average number of planets per system that could potentially support life.
- $\mathbf{f_l}$ = The fraction of habitable planets on which life actually emerges.
- $\mathbf{f_i}$ = The fraction of life-bearing planets where intelligent life evolves.
- $\mathbf{f_c}$ = The fraction of intelligent civilizations that develop detectable communication technologies.
- **L** = The average length of time such civilizations remain detectable.

The first three factors are becoming increasingly well understood thanks to modern astronomy. We now know that planets are common throughout the galaxy, and many stars host Earth-sized worlds within their habitable zones. However, the remaining factors remain largely unknown. We still do not know how easily life begins, how often intelligence evolves, or how long technological civilizations survive.

The Drake Equation therefore does not provide a definitive answer to the question of alien life. Instead, it serves as a framework for scientific discussion, highlighting both what we know and what remains one of humanity's greatest mysteries. Depending on the values assigned to its variables, the equation can predict anything from a galaxy teeming with civilizations to one in which humanity is effectively alone.



An illustration of the Earth-size planet TOI 700e orbiting within the habitable zone of its star.

So Close Yet So Far Away

The search for extraterrestrial life continues with the progress of science and astronomy. The search for extraterrestrial life is ultimately a search for our place in the universe. While life on Earth is carbon-based and depends heavily on liquid water, there is no guarantee that all life in the cosmos follows the same blueprint. The clues for alien existence continue to increase with each passing year due to the growing database of discovered Earth like planets and more theories suggesting alien life. We are so close yet so far away from a milestone in astronomical survey, and scientific progress in general. Some scientists have even speculated about alternative

forms of life based on different chemistries, such as silicon-based organisms or life existing in environments vastly different from our own. As our understanding of the universe grows, so does the possibility that life may be stranger and more diverse than we can currently imagine. Whether we are alone or merely one example among countless others remains one of humanity's greatest unanswered questions.

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